



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 4 • STANDARD 6.AT.4

# Understand Percent

Lesson 4-1 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

**Language Objective:** I can explain what a percent means using the words per hundred, part, whole, and equivalent ratio.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Percent

Per hundred

Decimal grid

Part-to-whole

Greater than 100%



Tap any word to see its meaning and a picture.

1

The word that means “per hundred” is .

2

The meaning hidden inside the word percent is   
.

3

A 10-by-10 grid of 100 squares used to show a percent is a  
 .

4

A comparison of one part against the whole amount is a  
 comparison.

5

A percent above 100 means the amount is  than one whole.

## Write About the Math

**What percent of the gray kitten's weight does the orange kitten weigh, and how does the tape diagram show it?**

*¿Qué porcentaje del peso del gatito gris pesa el gatito naranja, y cómo lo muestra el diagrama de cinta?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Percent**  
*Porcentaje*

☐ **Per hundred**  
*Por cada cien*

☐ **Decimal grid**  
*Cuadrícula decimal*

☐ **Part-to-whole**  
*Parte a todo*

☐ **Greater than 100%**  
*Mayor que 100 %*

**Model:** 150% is more than one whole, and 0.5% is a tiny piece smaller than 1%. — Students explain 150% is more than one whole (the player beat the goal) and 0.5% is far less than one whole (a tiny boost), connecting to the idea of 100% as the whole.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The gray kitten is the \_\_\_\_, so its weight is \_\_\_\_%. The orange kitten needs one whole tape plus \_\_\_\_ of another. Adding the parts, \_\_\_\_% + 50% = \_\_\_\_%. A percent greater than 100% means the orange kitten weighs \_\_\_\_ than one whole gray kitten.
- My answer is \_\_\_\_ because \_\_\_\_.

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.  
*Mi afirmación es \_\_\_\_.*
- My evidence is \_\_\_\_, so \_\_\_\_.

*Mi evidencia es \_\_\_\_\_, entonces \_\_\_\_\_.*

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### **Explain**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Percent
2. **Notes 2:** Per hundred
3. **Notes 3:** Decimal grid
4. **Notes 4:** Part-to-whole
5. **Notes 5:** Greater than 100%
6. **Watch for this mistake:** *A common mistake with percent is scaling only one term of the ratio: reading 7 out of 10 as 7%. A percent compares to 100, so BOTH terms must be multiplied by the same factor —  $10 \times 10 = 100$ , so  $7 \times 10 = 70$ , giving 70%. Reading it as 7% describes seats that are nearly empty when they are nearly full. A second mistake is treating a percent as an amount: 150% of a 200-point goal is  $1.5 \times 200 = 300$  points, not 150 points.*
7. **Try It 1:** D. Twice as much as you need — 100% is what you need, so 200% is two of those — twice as much.
8. **Try It 2:** — Dividing a percent by 100 tells you how many wholes it is.